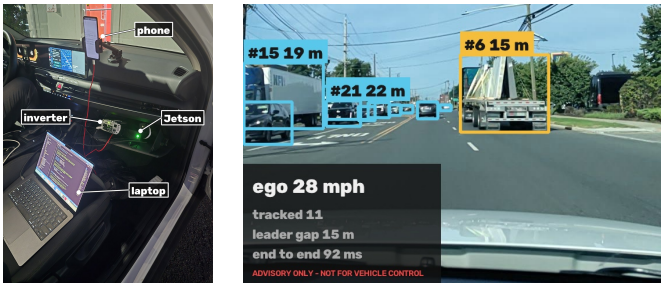
A Decentralized Speed Advisory System with Global Traffic Awareness and Local Visual Sensing

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(a) The rig. (b) The local layer's view on a drive.

Fig. 1. The deployed system. In (a), the phone supplies the camera, GPS and inertial unit and carries the display; the Jetson runs perception, both advisory layers and the safety filter; the inverter is its only connection to the car. In (b), the leader in the ego lane is amber and the other tracked vehicles are cyan. Nothing connects to the throttle, the brakes, the steering or the vehicle bus.

***Abstract*—Traffic jams can emerge rapidly when local disturbances push a road beyond its critical density, making early detection and intervention essential. Preventing such breakdowns without roadside infrastructure or centralized vehicle coordination is challenging: commercial traffic feeds provide network-wide information but are delayed and aggregated, while onboard sensors provide immediate observations over a limited range. We present a decentralized speed advisory system that combines these complementary sources through a global-local architecture. Each equipped vehicle independently computes a network-wide speed plan from a common, versioned traffic feed, enabling consistent advisories without vehicle-to-vehicle communication or a central speed coordinator. An onboard camera estimates local traffic occupancy and applies a bounded correction when the road ahead approaches its critical density. We analyze the timing limitations of delayed traffic observations and the sampling requirements for anticipatory local intervention. We implement the system on approximately US$500 of commodity hardware and evaluate its sensing, computation, and communication pipeline through public-road drives, using 18,818 recorded frames to construct a simulation camera model. In simulations on the Mainz highway network across five traffic disturbances, the system prevents jams in 55 of the 74 triggered runs that would otherwise jam, reduces average jam time from 130.8 to 3.9 segment-minutes, increases throughput by 3.9%, and reduces total vehicle time in the network by 12.6%. These results demonstrate the feasibility of decentralized onboard speed advisory and its potential to mitigate traffic breakdown without dedicated roadside infrastructure.**

I. Introduction

A highway that carries traffic close to its capacity can jam without a crash or a fixed bottleneck. A short burst of inflow, or a vehicle stopped in one lane for a few minutes, pushes the local density past its critical value. The outflow of the road then falls, and the queue sustains itself, spiralling into a suddent traffic jam [1]. Such a jam also takes significantly longer to drain. We simulated such conditions on the Mainz road network [2] by blocking a lane at the exit for five minutes, and found that it increases the network's total time under jam to 137 segment-minutes. A jammed road near capacity serves less traffic than keeps arriving and the queue does not drain (Section IV). Such a jam has to be prevented rather than cleared: the warning, and the intervention, must come before the road breaks down.

Existing controllers obtain that warning from infrastructure. Variable speed limits and ramp meters hold a bottleneck below breakdown using loop detectors and overhead gantries [3], [4], and vehicle-based controllers send target speeds to equipped vehicles from a central planner that observes the traffic [5]. Most roads have neither the roadside equipment nor the coordinator. One important concern for such approaches is the cost to deployment. Installing and maintaining infrastructure costs public funds, and a central controller has maintanence and privacy costs. Adoption is the limiting factor in both cases, since adoption dictates how much public benefit gets derived from these technologies.

Our system was designed with this deployment cost and adoption in mind, with a minimum set of changes to an ordinary vehicle that can enable this advisory. We consider two sources of information: a commercial traffic probe feed for every segment of the network, and a dashboard camera. The feed is global but aggregate and late: it reports segment speeds, which barely change until a road is about to break down, commercial feeds report a change in speed a mean of 4.4 minutes after it happens [6], and the feed we use publishes once a minute and becomes readable 29 to 90 s after each publication. The camera is immediate but local: from the gaps to the vehicles ahead it measures density, which rises before speed falls, but only over the next 60 m. Thus, each modality covers what the other misses.

This paper presents a speed advisory system built on that complementarity (Fig. 2). An onboard computer in each vehicle runs two layers. The *global layer* turns each published version of the feed into a speed plan for the whole network; the plan is a deterministic function of the version, and every vehicle switches to it at a time the version itself sets, so all vehicles show the same plan without exchanging a message. The *local layer* estimates the density of the road ahead from the camera every 5 s and lowers its own vehicle's advised speed by one step below the plan when that density is about to reach its critical value, while the feed shows the network

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near capacity. The driver sees one number: the advised speed for the current road segment.

We built the system from a phone and Jetson Nano, a US$250 onboard computer, and deployed it in a car on public roads in central New Jersey. We then built the simulated camera from the frames recorded on those drives. In simulation on the Mainz highway network, extended by 3 km approach roads that pace the demand into it, the system prevented the jam in 55 of the 74 triggered runs in which no control jammed, cut jam time from 130.8 to 3.9 segment-minutes per run, and served 3.9% more traffic than no control with 12.6% less travel time in the network; a controller reading the same feed without the camera left 11.6 segment-minutes of jam and served 5.4% less traffic.

Contributions. We make four contributions.

- We design a global-local advisory architecture in which vehicles agree on a network-wide plan by reading the same feed version, with no messages between them and no server computing their speeds, and each vehicle's camera corrects the plan by at most one step (Sections III-A, III-B).
- We measure the feed's delivery process and analyse the timing of feed-only control: we derive a lag condition under which no feed-only controller can act before a jam, and a sampled-margin bound that sets the local layer's horizon (Section IV).
- We deploy the system on US$500 of commodity hardware, report its latency and failure modes, and build a camera error model from 18,818 road frames (Section V).
- We evaluate the system over five jam triggers against no control, a feed-only lookahead and each layer alone, with its cost when no jam would form and its sensitivity to compliance and driver response (Section VI).

II. Related Work

Holding a bottleneck below breakdown. Ramp metering [4] and variable speed limits [3] limit the flow that reaches a bottleneck, observing it through loop detectors and acting through roadside signals. Adding floating-car probe data and traffic camera video to such a scheme shortened jam detection and improved its speed limits in simulation [7], so a controller that learns of a jam sooner controls it better. Our system pursues the same goal without roadside equipment: the vehicles both sense and act.

Vehicles as actuators. Prior work has shown that a single controlled vehicle damps stop-and-go waves on a ring road [8], and reinforcement learning controllers have been shown to work for mixed-autonomy traffic [9]. Our controller lies in the same vein, except the traffic state is not given but is obtained from a commercial feed and the vehicle's own camera. Earlier work has also established that coarse-grained advisories can steer human drivers [10], and the CIRCLES project deployed about one hundred vehicles on an interstate under a centralized speed planner [5]. These studies motivate our choice of a speed advisory design over alternatives. Telematics applications already show drivers feedback on their phones, hence, our system advises through a phone and leaves the vehicle's controls to the driver.

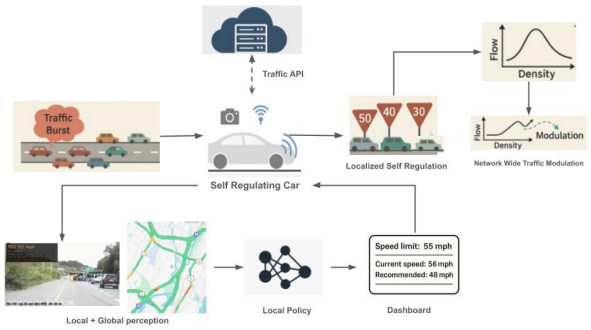


Fig. 2. System overview. A burst of inflow pushes a stretch of road toward its critical density. Each equipped vehicle reads the network from a commercial probe feed (global layer) and the road ahead from its own camera (local layer), computes its advisory onboard, and shows the driver a recommended speed. The vehicles coordinate only through the feed version they read; none talks to another.

Probe data and coordination without messages. For connected vehicles to share their own measurements over V2X, most vehicles would need to carry radios. The commercial probe feeds report a speed change minutes late [6]. To keep the decentralized design, our vehicles coordinate through common information [11], each reads the same feed version and computes the same plan from it. To mitigate the effects of lag, we use the dashboard camera to add a correction to the advisory.

III. Method

A. System Overview

Design goals. Five requirements shape the design. The first two are derived from the jam dynamics, and the last three are to lower the adoption cost.

(G1) *See the whole network.* A segment jams because of vehicles that entered upstream of it, so advice that prevents the jam must reach them before they arrive, from the state of segments they have not yet reached.

(G2) *Act before the road breaks down.* Near capacity a jammed bottleneck serves less traffic than keeps arriving, so the jam does not drain until demand falls (Section IV); a controller that acts while the road still flows can keep the jam from forming.

(G3) *Give consistent advice without a coordinator or messages between vehicles.* Vehicles on one stretch told different speeds change lanes and brake near capacity; a coordinator must track every vehicle and stay available, and messages need radios. We want coordination without that infrastructure.

(G4) *Advise without touching the vehicle's controls.* Taking over the throttle or brakes requires modifying and certifying the vehicle, as for the CIRCLES vehicles [5]; an advisory fits any car and leaves the driver in control.

(G5) *Run on hardware a driver can buy.* The system acts only through equipped vehicles, so the cost of equipping one limits how many take part; the rig of Section V costs US$500.

One layer per source. No single source meets G1 and G2 together (Table I): the feed covers the network but reports a forming jam late, and the camera reports it early but sees

TABLE I
THE TWO SOURCES OF INFORMATION, AS MEASURED IN THIS WORK.

	Feed (global layer)	Camera (local layer)
Coverage	every network segment	60 m, own and side lanes
Measures	segment speed, jam factor	gaps, vehicles in range
Update	one version per 60 s	every 5 s
Delay to act	99.5 s	about 6 s
Sees a jam when	speed has fallen	density is rising

Algorithm 1 Version-keyed plan, run on every vehicle

1: query the feed every P_q seconds, at the vehicle's own phase
2: on a response stamped T newer than the held one: compute the plan $a = \pi(x)$
3: apply a from $T + D$, or at once if $T + D$ has passed

only the next 60 m. The global layer therefore sets a plan for the whole network from the feed, and the local layer may lower one vehicle's speed below that plan when its own camera sees the road filling. A safety filter follows both: it raises an advisory far below the surrounding traffic, since a much slower vehicle is itself an obstruction, and withholds it when the gap ahead, or the time to close it, falls below a minimum.

Inputs and outputs. The network is divided into super-segments, twelve on the network we evaluate. For each, the global layer reads three fields from the HERE Traffic API [12], `speed`, `free_flow` and `jam_factor`, and two from the map, lane count and length. HERE does not publish how it computes the jam factor, so from the patent that defines it by travel time relative to free-flow travel time [13] we use a simple model, $10(1 - v/v_f)$ clipped to $[0, 10]$, for speed v and free-flow speed v_f. The output for each super-segment is 0.5, 0.75 or 1.0 of its static speed limit, not of the feed's free-flow reading, so that a live reading is not both observation and base.

B. The Global-Local Speed Advisory

Global layer: a plan every vehicle computes the same way. Each vehicle runs the same policy π on each published version of the feed, twelve super-segments with five fields each, and obtains one speed fraction per super-segment. We trained π in simulation by Q-learning with the action set and reward of a central controller that read the full traffic state [2]: the reward pays for speed and penalises density above the critical value, refitted to our vehicles. Because π is deterministic, vehicles that read the same version compute the same plan without communicating [11]. They do not read it at the same moment, because each vehicle queries on its own schedule, so each switches to a version at a time the version itself sets (Algorithm 1). With D at least the largest delivery lag plus the query interval, every vehicle holds a version before its switch time. Querying at a common second also avoids disagreement, but a version that becomes readable during those requests, or a missed query, splits the vehicles for a minute, and the publication phase moved by 26 to 34 s in our measurements.

Local layer: a one-step correction from the camera. Every 5 s the local layer estimates the occupancy $\hat{k}$ of the road ahead, the fraction of lane length that vehicles cover: in its own lane from the gap s to the leader, as $L/(s + L)$ with $L = 4.5$ m, and in each lane beside it from the number n of vehicles seen within the camera's effective range R, as nL/R; $\hat{k}$ is the mean over the visible lanes. The layer fits the trend $\dot{k}$ of $\hat{k}$ over the last 30 s and computes the margin to the critical occupancy k_c,

$$M = \big(k_c - \hat{k} - \tau \max(\dot{k}, 0)\big)/k_c, \tag{1}$$

with $k_c = 0.178$ and horizon $\tau = 10$ s. When $M \leq 0$ the vehicle shows the next lower speed fraction instead of the plan's. The correction holds for at least 30 s and releases once $M \geq 0.05$ on two consecutive samples, so one noisy frame cannot change the advice. Following Proposition 2 (§IV), τ is the layer's own response time, the sampling interval plus processing and the driver's reaction, rather than a tuned parameter.

When the correction may act. The camera sees only the next 60 m, and near capacity ordinary traffic also brings the occupancy ahead close to k_c, so a correction without a gate slows vehicles that did not need slowing. A feed gate therefore lets the correction switch on only while the feed version the vehicle applies shows a jam factor of at least $J_g = 1$ on some super-segment, measured against one step below that version's plan so that the advisory's own slowing does not read as congestion. Every vehicle that applies the same version sees the same gate, and a corrected vehicle is released through the hysteresis when its gate closes.

Why the correction only lowers, and by one step. A vehicle that sees an empty road ahead does not know that the network downstream is filling, which the plan does know, so it must not speed up; a vehicle that sees a filling road needs only to slow its own arrival, and one step bounds how far it falls below its neighbours and prevents obstructing the traffic flow from the vehicle itself. Vehicles on a segment thus differ from the shared plan by at most one step, and only where their cameras see the road filling.

IV. TIMING ANALYSIS OF JAM PREVENTION

The feed publishes once a minute and arrives tens of seconds late. Every response carries a stamp, `sourceUpdated`, the time its version was published; we polled the feed every 5 s for 1.9 hours over Mainz and over the area of our drives. Over Mainz, 113 versions arrived 60 s apart and each became readable 34.4 to 89.5 s after its stamp; over New Jersey the publication phase shifted, so versions arrived as little as 28 s apart, each readable 28.9 to 84.0 s after its stamp. The 123 responses the rig received on the road were 34 to 110 s old. A vehicle that keys its plan to the feed version (§ III-B) applies each version $D = \bar{L} + P_q$ after its stamp, where $\bar{L}$ is the largest lag and P_q the query interval: 99.5 s on the Mainz process. This delivery delay adds to the feed's own probe latency [6].

The lag condition. Let t_v be the first time a segment's speed departs from what it would have been without the triggering event, t_{on} the onset of the jam, and $W = t_{\text{on}} - t_v$ the warning the speed signal gives. Every version stamped before t_v is the same with and without the event.

Proposition 1 (Lag condition)**.** *With publications at most g seconds apart and each version applied D seconds after its stamp, a feed-only controller changes no advised speed before $t_v + D$, and so changes none before onset if $W < D$. If $D \le W < D + g$ and the stamp phase is uniform, it can act before onset with probability $(W - D)/g$.*

Proof sketch. The first stamp at or after t_v falls in $[t_v, t_v + g)$ and is applied in $[t_v + D, t_v + D + g)$; nothing applied earlier differs from the case without the event. Acting before onset needs that stamp in $[t_v, t_{\mathrm{on}} - D)$, an interval of length $W - D$ out of g. □

The condition is necessary, not sufficient, because a changed advisory still has to change how vehicles drive before the jam forms. Speed makes W short: on the free-flow branch of the speed-density relation, speed hardly changes as density rises until the road breaks down, while density, which a camera measures from the gaps to the vehicles ahead, rises from the start.

How often the local layer must sample. A rule that observes density only at sample instants protects the critical density only if it acts early enough to absorb the rise that can happen before its next action.

Proposition 2 (Sampled margin)**.** *Let local density rise no faster than r, let a rule sample it every Δ and act l after a sample, and let its action stop the rise. If the rule acts at the first sample at or above $k^* - r(\Delta + l)$, density stays below k^*.*

Proof sketch. The sample before the one that triggers the action was below $k^* - r(\Delta + l)$, and density rises by at most $r(\Delta + l)$ before the action. □

For the feed, $\Delta + l$ is $60\,\mathrm{s} + D$, about 160 s on the Mainz process; for a camera sampled every 5 s with about 1 s of processing it is 6 s. The local layer's margin is therefore over 25 times smaller, so it can act close to the critical density.

Why preventing a jam is worth more than draining it. A jam behaves like a queue at the bottleneck, and the queue's arithmetic decides whether it ever clears.

Proposition 3 (Point-queue drain)**.** *A queue of Q vehicles with arrival rate λ and service rate μ empties after $Q/(\mu - \lambda)$ if $\mu > \lambda$ and never otherwise. If the bottleneck serves μ_f before it breaks down and $\mu_q < \mu_f$ after, and $\mu_q \le \lambda < \mu_f$, no queue forms while it has not broken down, and none drains once it has.*

On the Mainz network the jammed exit serves 3,535 veh/h under no control, 37 veh/h less than the background demand of 3,572 veh/h, so a jam that forms there never drains, and preventing the breakdown is worth more than draining the queue faster.

V. Implementation and On-Road Deployment

Hardware. The rig is built from commodity parts (Fig. 1): a mid-range Android phone (OnePlus Nord N10, US$150), an NVIDIA Jetson Orin Nano developer kit (US$250), and US$100 of mount, 12 V inverter and cable, US$500 in total. Nothing connects to the vehicle's controls or its bus, and no server takes part in computing an advisory: the only network traffic is the feed query.

Software. The phone captures the camera, GPS and inertial unit, queries the feed once a minute, and forwards everything raw over USB; the Jetson runs detection (YOLOv8n at 640 px), tracking, distance and density estimation, both advisory layers, the safety filter, and a scheduler that sets each sensor's sampling rate. Interpretation runs on the Jetson because computing on the phone as well would worsen the thermal problem below.

Drives. We drove the rig on 2026-09-08 on public roads in central New Jersey. Eight runs recorded 22,929 ticks over 152.8 minutes, with a valid GPS fix on 99.1% of them, and 123 feed responses, every one returning HTTP 200. In two runs the phone applied the sampling rates the scheduler sent, and in six it only logged them. No driver acted on an advisory: that needs institutional review board approval, which this project does not hold, and one instrumented vehicle cannot change how traffic flows, so the traffic results are simulated (§ VI).

Computation is not the bottleneck; heat is. On the Jetson the path from detection to the decoded advisory takes a median of 19.8 ms, and the phone-to-Jetson link has a 95th-percentile latency of 116.19 ms over USB against 215.63 ms over a network link, which is why the rig uses USB. The failures were physical. On the first drive the phone reached Android's severe thermal tier on a windscreen mount in sunlight and the platform closed the application; a failed cable and a hotspot shut off by sunlight cost two further drives.

From recorded frames to the simulated camera. We ran the rig's own detector, tracker and distance code on the 18,818 frames that six drives indexed against the other sensors; the simulated camera uses the error tables of the four drives recorded at five frames per second. Measured against a second pass at 1,280 px, which sees farther, the detection rate falls from 0.95 within 10 m to 0.53 at 40 to 50 m and 0.32 at 50 to 60 m, which sets the range at 60 m. We compute distance from each box's width and an assumed vehicle width. Distance carries the frame-to-frame noise we measured, from 0.05 m at 10 m to 1.1 m at 50 to 60 m, and a per-vehicle offset with a spread of 31%, because a width-based distance is wrong by as much as that vehicle's width differs from the assumed one. The simulated camera sees its own and the adjacent lanes, with nearer vehicles hiding those behind them. The phone's focal length was not recorded, so the absolute distance scale is assumed (§ VI-F varies it).

VI. Evaluation in Simulation

A. Testbed and Protocol

Network. We use the model of the Mainz highway network from [2] in SUMO. It has twelve super-segments and ends in an exit where we made three lanes become two, which is where it breaks down. In a simulation a burst appears at the entrances out of nowhere, while on a real road its vehicles arrive along the roads upstream, so we added a 3 km approach road before each of the eight entrances. The approach roads pace the demand into the network, and nothing on them is measured. Controlled vehicles follow the advice through SUMO's speed

TABLE II
ABLATION OVER SEEDS 16–30: FIVE TRIGGERS (75 RUNS) AND NO TRIGGER (15 RUNS). JAM-FREE: RUNS WITH NO JAM. JAM: JAMMED SEGMENT-MINUTES PER RUN. THROUGHPUT (VEH/H) AND TRAVEL TIME IN THE NETWORK (VEH-H): ABSOLUTE FOR NO CONTROL, OTHERWISE THE CHANGE AGAINST IT ON THE SAME SEEDS. NO GATE AND TRUE OCCUPANCY: THE GLOBAL-LOCAL ADVISORY WITHOUT ITS GATE, AND WITH THE CAMERA REPLACED BY THE TRUE OCCUPANCY IN ITS WINDOW.

	Five triggers				No trigger
Controller	Jam-free	Jam	Throughput	Travel time	Throughput
No control	1	130.8	3,456	1,622	3,546
Feed lookahead	46	11.6	−5.4%	+11.9%	−20.8%
Plan only	39	22.1	+2.6%	−10.0%	−2.9%
Camera only	5	95.5	+1.3%	−1.1%	−0.9%
No gate	51	4.4	+2.5%	−6.0%	−6.9%
Global-local	**56**	**3.9**	**+3.9%**	**−12.6%**	**−3.0%**
True occupancy	58	3.4	+4.3%	−14.3%	−3.0%

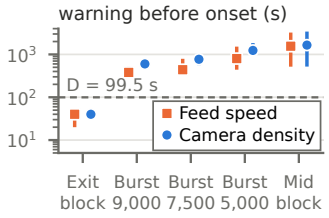


Fig. 3. How early each signal warns: time before the first jam of each no-control run at which the feed's speed fell below 80% of its pre-trigger level, or the camera's density rose three standard deviations above its own, for 30 s. Medians and interquartile ranges; dashed: the feed delay D.

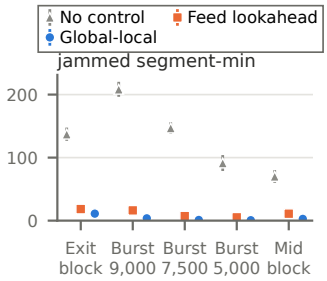


Fig. 4. Jam time by trigger, ordered by how soon no control jams: jammed segment-minutes, mean and two standard errors over 15 seeds.

cap, so car-following still brakes for the vehicle ahead; none of the 3,000 controlled runs had a collision.

Triggers. Every run starts from a background demand of 3,572 veh/h, which lies between the flow a jammed exit serves and the flow at which a free one breaks down, so the network does not jam by itself and a jam that forms does not drain (Proposition 3). Fifteen minutes in, one of five triggers is applied. Three are bursts that raise the total inflow to 5,000 veh/h for ten minutes, or to 7,500 or 9,000 veh/h for five; under no control the first jam starts a median of 14 to 28 minutes after the burst begins. Two are stopped vehicles that block a road for five minutes: one lane of the exit road, which jams the exit a median of 80 s later, and two lanes mid-network, which jams it a median of 37 minutes later. Each run lasts two hours, and the feed replays the polled Mainz process with $D = 99.5$ s.

Metrics. A super-segment is jammed when its mean speed is below 19 km/h, a threshold fitted to the network's speed distribution by the rule of [1], and a jam starts when that state lasts 120 s. Jam time is the jammed segment-minutes on the twelve super-segments of the network, and a run is jam-free when the network has no jam. Throughput is the flow served at the exit. Travel time is the vehicle-hours all vehicles spend in the network.

Baselines, ablations and protocol. No control lets every vehicle drive at its own desired speed. The feed lookahead is the controller one would build from the feed alone: it advises 0.5 of the limit when the next super-segment downstream has a jam factor of 2 or more, and 0.75 when the one after it has. The ablations each remove one part: the camera (plan only), the plan and the feed (camera only, on a plan of 1.0), the gate, and the camera's error (true occupancy in its 60 m window). All run at full penetration on the same 15 evaluation seeds. Thresholds were chosen on five separate selection seeds, the gate last, as the candidate with the least jam among those costing at most 1% of throughput beyond the plan alone with no trigger. We report means over seeds and, for differences, the paired mean with two standard errors.

B. The Camera Warns Before the Feed

On the three bursts the camera's density warning came first on 43 of 45 runs, by a median of 210 to 480 s (Fig. 3). The reason is the shape of the speed-density relation: as occupancy rises to 0.05 the median segment speed falls only from 57.5 to 55.0 km/h, and then to 35.8 km/h by 0.07. On the exit block the feed's warning was shorter than D on all 15 runs, so by Proposition 1 no controller that reads only the feed could change a speed before that jam began.

C. Jams Prevented and Jam Time

No control jammed on 74 of the 75 triggered runs. The system was jam-free on 56, preventing 55 of those jams, and cut the mean jam time from 130.8 to 3.9 segment-minutes per run, by 97% (Table II, Fig. 4). After the exit block, which jams within 80 s, it prevented no jam but shortened it from 137.1 to 11.1 segment-minutes. Against the feed lookahead it left 67% less jam (-7.8 ± 1.3 segment-minutes), less after every trigger, and served 9.9% more traffic.

D. What Each Layer Contributes

Neither layer prevents the jams alone (Table II). The plan alone was jam-free on 39 runs and on none after the largest burst, which the feed reports late. The camera alone was jam-free on 5, although it changed the shown speed on 32% of vehicle-ticks: from a plan of 1.0 its one step reaches only 0.75 of the limit. Together they were jam-free on 51 runs without the gate and 56 with it, and true occupancy in place of the camera added two more. The system served 3.9% more traffic than no control (±0.3%) with 12.6% less time in the network (±1.4%); the lookahead served 5.4% less, because it slows every vehicle upstream of a segment the feed reports as slowing, whether or not its own road is filling.

E. The Cost When No Jam Would Form

Near capacity, ordinary traffic also brings the occupancy ahead close to k_c. With no trigger, where no controller jams, the layers without the gate served 6.9% less traffic than no control, against the plan alone's 2.9%, and 5.9% less on true occupancy, so the loss comes mostly from the rule rather than from the camera's error. The gate cut it to 3.0%, and the share of vehicle-ticks with a corrected speed from 15.5% to 0.2%. Letting the camera also open the gate, at $\hat{k} \geq 0.8k_c$ or k_c,

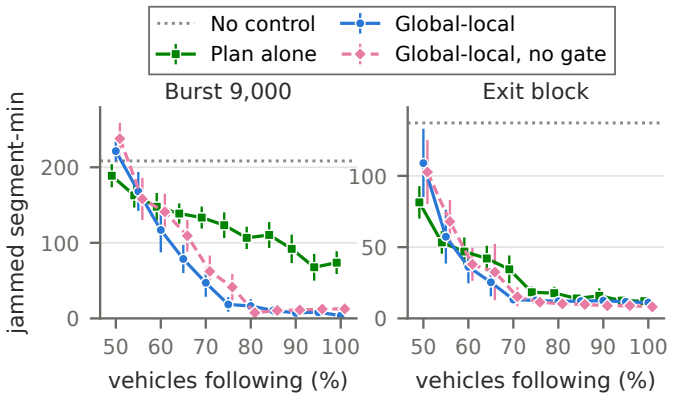


Fig. 5. Jam time against the share of vehicles that follow the advice, on the largest burst and the exit block: jammed segment-minutes, mean and two standard errors over 15 seeds. No control is the dotted line.

exceeded the 1% bound on the selection seeds, and releasing the correction while the road ahead discharges changed jam by less than the tie margin. The gate's price is the exit block, which jams before the feed opens the gate: 3.0 ± 0.9 more segment-minutes than without it.

F. Partial Compliance and Driver Response

Share of drivers who follow. On the largest burst and the exit block (Fig. 5), the system left less jam than no control at every share on the exit block and from 55% on the burst, and at 75% it left 18.5 segment-minutes after the burst against 41.4 without the gate. At 50% on the burst it left 13.0 ± 15.9 more than no control, against 29.5 ± 24.6 without the gate, and 32.6 ± 17.2 more than the plan alone; releasing the correction in a discharging queue did not reduce this ($+13.1 \pm 21.3$).

Driver response. We placed a driver between the display and the speed cap, who reacts to a change after a lognormal reaction time, aims above the advice by a compliance offset, and slows toward a lower advice at a bounded deceleration. The parameters are assumed, since no driver acted on an advisory on the drives: a median of 2 s, $5 \pm 5\%$ and 1.5 m/s^2 (nominal), and 4 s, $15 \pm 10\%$ and 1.0 m/s^2 (lax). The system then left 11.1 and 46.7 segment-minutes after the burst (3.7 with exact compliance) and 12.3 and 13.7 after the exit block (11.1), less than no control on every seed and than the plan alone under the same drivers.

Camera scale and feed delay. With camera distances scaled by 0.8 and 1.25 (§ V), the system's jam changed by at most 1.8 segment-minutes and stayed below the lookahead's; without the gate the burst at 0.8 left 25.0 more than the lookahead (± 15.9). With D of 0, 189 and 368 s it left less jam than the lookahead on the exit block at every delay and on the burst at 0 s. At 189 and 368 s on the burst the two were within noise ($+2.9 \pm 10.8$, -0.8 ± 5.3), because the gate also waits for the feed: the layers left 4.9 and 6.8 segment-minutes without it, 14.5 and 17.5 with it.

VII. Conclusion and Limitations

A commercial probe feed and a dashboard camera, combined in a global-local advisory, let ordinary vehicles prevent jams that the feed alone reports too late: the feed coordinates every vehicle on one network-wide plan without messages and tells the camera's correction when the network is near breakdown, and the camera lowers one vehicle's speed minutes before the feed's speed would warn. In simulation the system, which runs on US$500 of commodity hardware, prevented the jam in 55 of the 74 triggered runs in which no control jammed and cut jam time by 97% while serving 3.9% more traffic than no control; neither layer did this alone.

Limitations. No driver acted on an advisory, so the traffic results are simulated on one network with assumed driver responses, and with half of the vehicles following, the system left more jam than the plan alone after the largest burst. The gate costs jam after a stopped vehicle that jams the exit faster than the feed reports, and it makes the result depend on the feed's delay. The camera's distance scale is assumed, and the feed's delivery process was measured on one day.

Future work. We will put the advisory in front of drivers, first on a closed course and then in a small fleet; look for a camera signal that separates a forming jam from ordinary traffic near capacity, so that the gate need not wait for the feed; fit the camera's distance scale against a ranging sensor; and poll the feed over several weeks.

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